

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Concept Mapping
Weightage:	35%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	ECE	Date:	16-07-2026

Instructions to Students

- Identify the key computer architecture concepts, components, and performance parameters mentioned in the given topic or scenario.
- Organize the identified concepts logically by establishing relationships between CPU, memory, I/O devices, bus structures, instruction execution cycle, and performance metrics.
- Represent the concepts using an appropriate concept map with clear nodes, connecting lines, and meaningful linking phrases.
- Demonstrate the hierarchy and interdependence among concepts, showing how one concept influences or interacts with another.
- Ensure the concept map is complete, accurate, well-structured, and easy to understand by using appropriate terminology and logical connections.

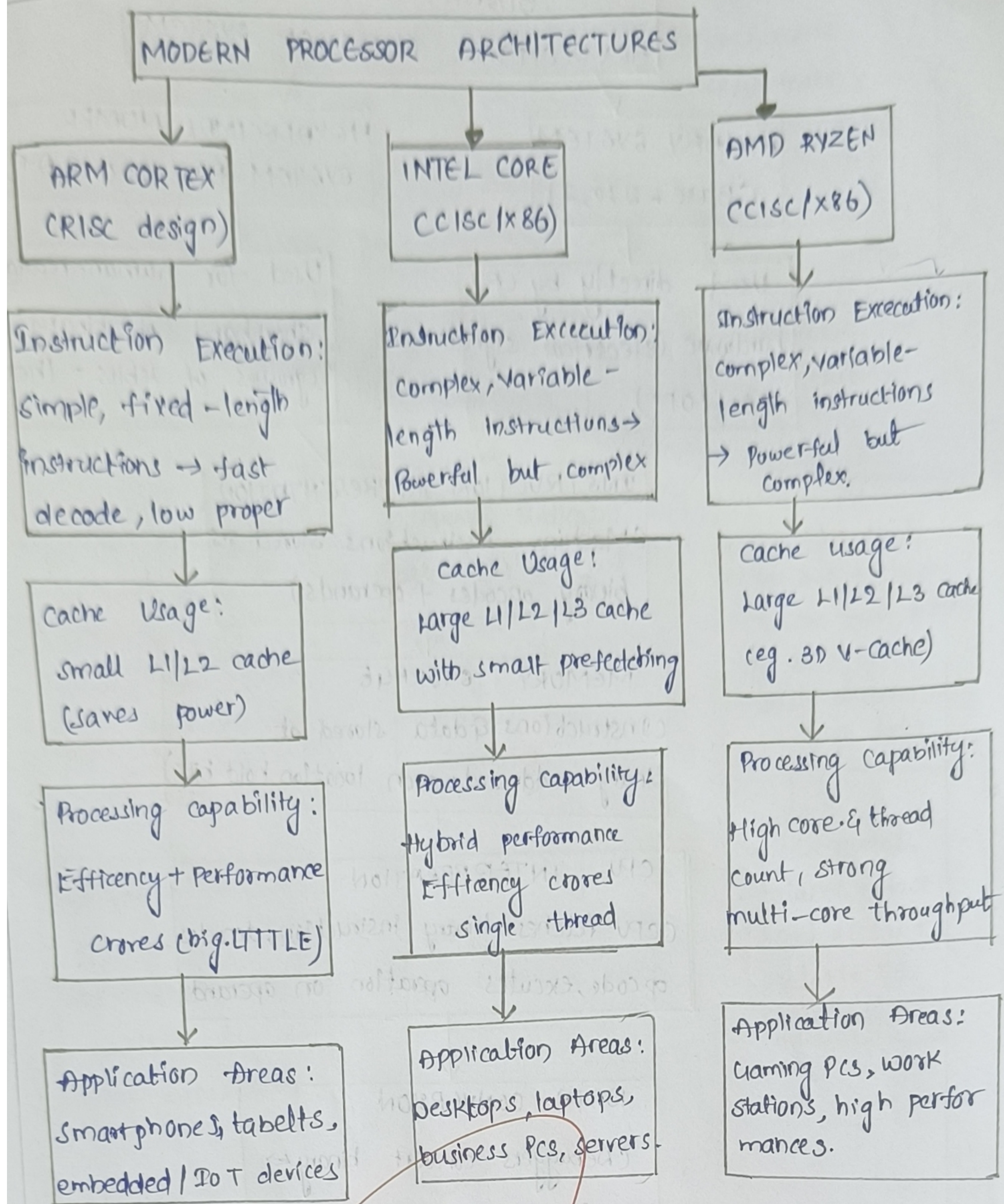
QUESTIONS

1. A software company is upgrading its office computers and must decide whether to use ARM Cortex, Intel Core, or AMD Ryzen processors for different workloads. Construct a concept map linking these modern architectures with their instruction execution, cache usage, processing capability, and application areas. Show how their architectural features influence performance and energy efficiency. How do modern processor architectures differ in meeting the requirements of mobile, desktop, and high-performance computing systems?
2. A computer technician is troubleshooting a system that stores machine instructions in memory but displays memory addresses in a debugger. Construct a concept map linking Binary and Hexadecimal Number Systems with instruction representation, memory storage, CPU interpretation, and data conversion. Show how instructions are represented, stored, and interpreted during execution. How do binary and hexadecimal number systems simplify memory organization, instruction representation, and debugging?
3. A manufacturing company is upgrading its automated inspection system and notices delays when multiple sensors and storage devices communicate with the processor simultaneously. Construct a concept map linking CPU, Memory, I/O Devices, Single-bus, Multi-bus, and Hierarchical Bus structures, showing how data and control signals are transferred. Illustrate the effect of each bus organization on communication efficiency. How do different bus structures influence system performance and reduce communication bottlenecks?
4. A programmer is developing an industrial automation application and must choose the most suitable instruction format and addressing mode for faster execution and efficient memory access. Construct a concept map linking Instruction Formats, Immediate, Direct, Register, Indirect, and Indexed Addressing Modes with Data Transfer, Arithmetic, and Control Instructions. Show how each addressing mode affects execution speed and memory access. How do instruction formats and addressing modes influence program execution and processor efficiency?
5. An embedded systems engineer is comparing the 8085 and 8086 microprocessors for a factory monitoring system that requires efficient instruction execution and memory organization. Construct a concept map linking their instruction sets, register organization, addressing modes, and memory organization, including memory segmentation in 8086. Show how these features affect program execution and system performance. How do the architectural differences between the 8085 and 8086 influence execution efficiency and memory management?

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____



2)

Concept Map - Binary and Hexadecimal number system

BINARY AND
HEXADECIMAL NUMBER SYSTEMSBINARY SYSTEM
(Base-2: 0, 1)HEXADECIMAL NUMBER
SYSTEM (Base-16: 0-9, A-F)Used directly by CPU
hardware (electrical signals
: ON/OFF)Used for human-read
shorthand of binary
(groups of 4 bits = 1 hex)INSTRUCTION REPRESENTATION
(Machine instructions stored as
binary opcodes + operands)MEMORY STORAGE
(Instructions & data stored at
binary address; each location holds bits)CPU INTERPRETATION
(CPU fetches binary instruction; decodes
opcode, executes operation on operands)DATA CONVERSION
(Debuggers convert binary $\leftrightarrow$
hex for easier reading; hex is
compact shorthand)

PU, MEMORY, I/O DEVICES ↔ BUS STRUCTURES

CPU
(issues, requests,
control signals)

MEMORY
(stores data &
instructions)

I/O Devices
(sensors, storage
devices, actuators)

Communicate via shared BUS
(data BUS + address BUS + control BUS)

Single-BUS
STRUCTURE

MULTI-BUS
STRUCTURE

HIERARCHICAL
BUS STRUCTURE

One common bus,
shared by CPU,
Memory & I/O
→ only ONE device
can transfer data
at a time

Simple & low cost,
but causes delays/
bottlenecks when
multiple devices
need access to bus

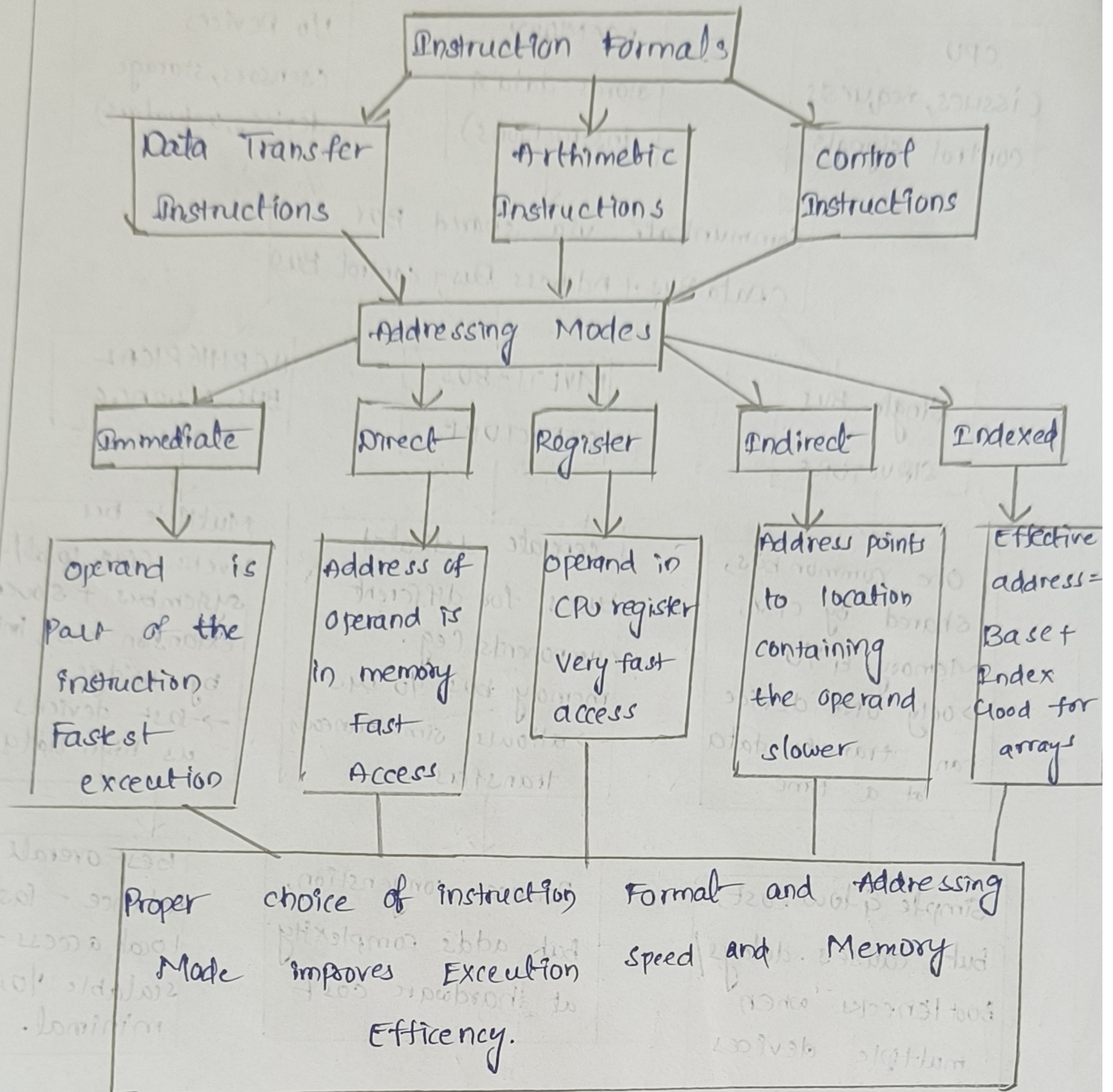
separate dedicated
buses for different
components (eg.
memory bus, I/O bus)
→ allows simultaneous
transfers

Reduces congestion,
but adds complexity
at hardware cost

Multiple bus
levels (fast local/
system bus + slower
expansion bus), linked
by bridges
→ fast devices
use high data.

Best overall
balance - fast
local access +
scalable I/O,
minimal.

4) Instruction Formats and Addressing Modes.



Comparison of 8085 and 8086 Microprocessors

